

# FD-VAD: SEMANTIC ENDPOINT DETECTION FOR STREAMING FULL-DUPLEX SPEECH

Puneet Mathur, Dinesh Manocha

University of Maryland, College Park

## ABSTRACT

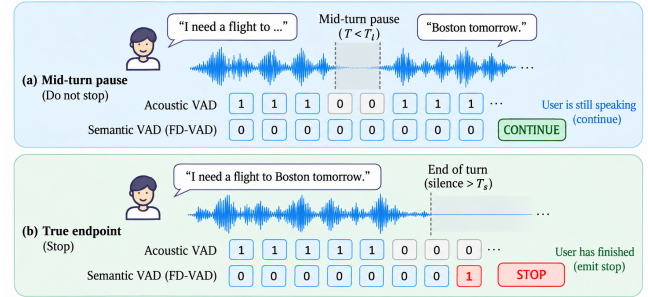
Natural turn-taking in full-duplex voice interaction requires determining from partial speech whether a pause reflects hesitation or a completed conversational intent. Acoustic voice activity detection lacks this semantic information, while cascaded ASR-based endpointing introduces transcription dependence and additional processing stages. We formulate semantic endpoint detection as a *causal audio-language reasoning task* and introduce **FD-VAD**, an ASR-free streaming endpointer that maps bounded causal audio windows directly to CONTINUE/STOP decisions. FD-VAD combines a frozen speech encoder with a lightweight modality adapter and a parameter-efficiently adapted language model, using a last-chunk training objective for streaming inference. We further introduce confidence-gated endpoint commitment to control interruption versus delay and boundary-focused hard-negative sampling to improve decisions around ambiguous turn boundaries. Across in-domain and conversational evaluations, FD-VAD outperforms strong streaming and non-streaming semantic turn classifiers, and achieves the highest EOT recall among qualifying systems on TurnBench dev set 0.853 (at  $FP \leq 0.10$ ) in a zero-shot setting. These results show that semantic endpointing can be performed directly from streaming audio without intermediate ASR or dialogue state tracking.

**Index Terms**— semantic endpoint detection, Voice Activity Detection, turn taking, full duplex, streaming speech

## 1. INTRODUCTION

Natural spoken interaction depends not only on *what* is said, but also on *when* each participant chooses to speak. Humans continuously infer whether another speaker has completed a thought, is pausing to plan what to say next, or intends to continue after a hesitation. This ability is fundamental to fluid turn-taking: responding before a speaker has finished is disruptive, while waiting too long after a completed request makes an interaction feel unnatural. The problem becomes particularly important for full-duplex voice agents, which continuously listen and speak, and therefore cannot rely on the rigid listen-respond alternation used by conventional voice assistants [1, 2]. A full-duplex agent must instead maintain an evolving estimate of whether the user still holds the conversational floor.

Voice activity detection (VAD) is commonly used in spoken systems to locate acoustic speech boundaries. However, acoustic VAD is often insufficient because a period of silence does not necessarily indicate that a conversational turn has ended. A speaker may pause while searching for a word, producing a disfluency, or planning the remainder of a request; conversely, a short utterance may already express a complete intent. The decision is therefore inherently *semantic*: the system must determine whether the observed speech constitutes a complete conversational act, rather than merely detect whether acoustic energy has disappeared. We refer to this problem as *semantic endpoint detection*. In a streaming setting, it is espe-



**Fig. 1.** Streaming semantic endpoint detection task, where 0 and 1 denote CONTINUE and STOP, respectively. **FD-VAD** uses linguistic as well as acoustic cues for end-of-turn detection.

cially challenging because the decision must be updated from partial audio without access to future speech.

Existing endpointing approaches expose a trade-off between semantic reasoning, streaming operation, and architectural simplicity. Acoustic methods such as Silero VAD [3] are lightweight and naturally streaming but cannot determine whether the preceding speech is semantically complete. Cascaded approaches first transcribe speech and then use a language model or text classifier to determine whether the utterance is complete [4, 5]. These methods provide access to lexical semantics, but introduce an explicit ASR dependency into the turn-taking loop and inherit transcription errors and additional processing stages. Semantic approaches such as Smart Turn [6], Easy Turn [7], FastTurn [8], Next-Turn [9], and SpeculativeETD [10] incorporate semantics through utterance-level classification, transcript/CTC pathways, timing supervision, or silence-triggered semantic inference. At the other end, full-duplex systems such as Moshi and Freeze-Omni [1, 2] learn turn-taking jointly with response generation. These methods improve semantic awareness, but typically introduce intermediate dependencies, triggering mechanisms or couple to particular dialogue architectures.

We argue that semantic endpointing need not require these intermediate mechanisms and instead formulate it as a *streaming audio-language reasoning task*: given only the speech observed so far, determine whether the speaker has expressed a complete conversational intent. We introduce **FD-VAD**, an ASR-free semantic endpoint detector for streaming full-duplex speech. FD-VAD maps short causal waveform windows directly to CONTINUE/STOP decisions. We utilize a frozen self-supervised speech encoder extracts acoustic and linguistic representations from the incoming waveform, while a lightweight modality adapter maps them into the embedding space of a small language model that performs the semantic endpoint decision. Rather than processing the complete utterance, FD-VAD repeatedly evaluates a bounded sliding window and supervises only the state at its most recent chunk. The resulting model reasons about

semantic completeness while preserving causal, fixed-cost streaming inference. Importantly, FD-VAD requires neither transcript generation nor a separate acoustic-to-semantic cascade, and its endpointing policy remains independent of the downstream voice agents.

Additionally, we address two properties particularly important for real-time deployment. First, endpoint errors are asymmetric: an early STOP can interrupt a user, whereas a conservative decision primarily adds delay. We therefore introduce a **confidence-gated decision mechanism** that explicitly controls this trade-off through the predicted stop probability and the number of consecutive decisions required before committing an endpoint. Second, ambiguous examples are concentrated near the true turn boundary rather than uniformly throughout an utterance. To better model these cases, we introduce **boundary-focused hard-negative sampling**, which emphasizes difficult boundary regions during training while leaving the inference architecture unchanged. **Our main contributions are:**

- We reformulate semantic endpoint detection as a **causal audio-language reasoning task**, where the model determines from partial speech whether the user should retain or yield the conversational floor.
- We introduce **FD-VAD<sup>1</sup>**, an ASR-free audio-to-LLM architecture that performs streaming semantic endpoint detection for full-duplex speech interaction.
- We propose **confidence-gated endpointing** and **boundary-focused hard-negative sampling** to control the interruption-latency trade-off and improve decisions near true endpoints.
- FD-VAD achieves the **highest EOT recall on TurnBench dev** among compared systems (0.853 at  $FP \leq 0.10$ ) in a zero-shot setting, and matches a strong non-streaming turn classifier at the utterance level (0.965 vs. 0.966).

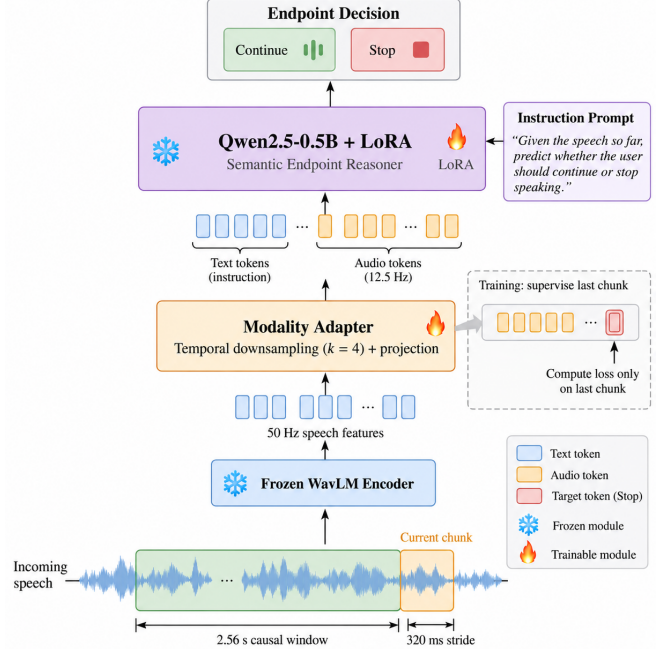
## 2. RELATED WORK

**Transcription- and segmentation-dependent endpointing.** Acoustic VAD detects speech boundaries from energy, spectral, or learned cues [3, 11], but provides no measure of semantic completion. Cascaded systems such as TEN [4] obtain semantics through an ASR transcript, while Smart Turn [6] predicts completeness directly from an acoustically segmented utterance. FD-VAD instead operates continuously on causal audio windows without requiring either transcription or a completed speech segment.

**Streaming methods with intermediate signals.** Recent methods reduce this dependence while retaining auxiliary mechanisms. Easy Turn [7] jointly predicts ASR and turn state, FastTurn [8] relies on streaming CTC representations, SpeculativeETD [10] triggers semantic inference after detecting silence, and Next-Turn [9] learns endpoint timing through time-to-next-speech supervision. FD-VAD removes these intermediate pathways and directly maps observed speech to a semantic CONTINUE/STOP state.

**Dialogue-coupled and speech-language models.** Full-duplex models such as Moshi and Freeze-Omni [1, 2] learn turn-taking jointly with response generation, while TurnFSM [12] embeds semantic VAD within an LLM-based dialogue-state controller. FD-VAD instead remains independent of the downstream voice agent. Architecturally, it follows the audio encoder–adapter–LLM paradigm used in speech-language models [13–15], specializing it for causal endpoint prediction using bounded audio context and a last-chunk objective.

<sup>1</sup>Code and model release: <https://fd-vad.github.io/>



**Fig. 2. FD-VAD architecture.** A causal audio window of incoming speech is encoded by frozen WavLM, projected into the embedding space of a LoRA-adapted Qwen2.5-0.5B language model for CONTINUE/STOP endpoint prediction. Training supervises only the final chunk of each window.

## 3. METHOD

**Architecture:** Figure 2 shows FD-VAD, an autoregressive LLM-based architecture for semantic endpoint detection, containing three components: (i) **WavLM Audio Encoder.** A frozen WavLM-based-plus encoder [16] maps raw waveform to frame-level speech features. (ii) **Audio-Text Modality Adapter.** To bridge the modality gap between audio and text, we utilize a modality adapter composed of two linear layers with a ReLU activation function that temporally downsamples and projects audio features into the embedding space of an LLM. (iii) **LLM Backbone.** We use Qwen2.5-0.5B-Instruct [17] as the LLM backbone, which takes the adapted audio features concatenated with text prompt embeddings as input to predict the semantic completeness state of the input speech segment. Let  $A^I$  denote the WavLM feature sequence. The adapter groups every  $k$  consecutive frames and applies  $A^P = W_2 \text{ReLU}(W_1 A^I + b_1) + b_2$ . We use  $k=4$ , so the 50 Hz speech representation becomes a 12.5 Hz sequence of audio tokens. The projected audio embeddings are concatenated with a short instruction prompt, and the LLM predicts a single state token, CONTINUE or STOP, followed by  $\langle \text{eos} \rangle$ . The speech encoder and base LLM remain frozen. We jointly optimize the modality adapter and low-rank LoRA parameters [18], for 10.3M trainable parameters (1.7% of the total parameter count). This keeps the endpoint detector modular and parameter-efficient while reusing strong pretrained speech and language representations.

**Causal Sliding-window Inference:** To enable streaming inference, we employ a sliding window training strategy that makes predictions using only the audio within each window, reducing dependence on the entire input sequence, enabling incremental chunk-wise predic-

tions, and offering latency advantages. FD-VAD operates on a 10 ms waveform slicing grid. For chunk index  $c$ , the model advances by a stride of  $S_f=320$  ms and retains at most  $W_f=2.56$  s of left context. Thus, each decision is made every 320 ms from a 2.56 s causal window which is encoded by WavLM into approximately 128 feature frames at 50 Hz. The adapter downsamples these by  $k=4$  to approximately 32 audio tokens at 12.5 Hz before the LLM.

**Last-chunk Training Objective:** For each window, supervision is applied only to its terminal chunk. Let  $y_c \in \{0,1\}$  denote CONTINUE and STOP, respectively, and define the target sequence  $Y_c=\{y_c, \langle \text{eos} \rangle\}$ . We minimize the autoregressive cross-entropy

$$\mathcal{L}(\theta) = - \sum_{t=1}^{|Y_c|} \log p_{\theta}(Y_c^{(t)} | Y_c^{(<t)}, A_c^P, T^P), \quad (1)$$

where  $A_c^P$  denotes the adapted audio embeddings for the current window and  $T^P$  is the instruction prompt.

**Confidence Gating Commitment:** A single erroneous STOP can prematurely interrupt a user. At inference time we therefore optionally commit STOP only when  $P(\text{STOP}) \geq \tau$  for  $K$  consecutive decisions. Increasing  $\tau$  or  $K$  reduces false interruptions but also delays the commit. We treat this confidence gate as an operating-point controller layered on top of the base model.

**Boundary-focused Hard-negative Sampling:** Training errors concentrate near the true endpoint, where a one-chunk timing shift changes the label. We consider boundary-focused hard-negative sampling that oversamples windows whose terminal chunk lies within  $\pm 2$  chunks of the endpoint. This leaves inference unchanged but reallocates training mass towards ambiguous boundary cases.

#### 4. DATA AND EXPERIMENTAL SETUP

**Training Data:** We use the English subset of smart-turn-v3.1 [19], a 16 kHz conversational endpointing corpus with native *complete* and *incomplete* utterance labels. We follow the official train/test partition. The training set contains 63,386 clips (31,338 complete / 32,048 incomplete; 156 h), comprising 42,529 human and 20,857 TTS recordings. For complete utterances, the STOP transition is placed at the acoustic speech offset ( $-40$  dB relative to peak), with subsequent chunks labeled STOP; incomplete utterances remain CONTINUE throughout. Incomplete examples are corpus-native rather than artificially truncated complete utterances.

**Training Hyperparameters:** We freeze the WavLM encoder and LLM backbone and train only the modality adapter and LoRA parameters ( $r=16$ ) for one epoch using AdamW with learning rate  $2 \times 10^{-4}$ , cosine decay, and 3% warmup. Training uses an effective batch size of 64 in bf16 on two A100 GPUs.

**Evaluation Data and Protocols:** The held-out in-domain test set contains 1,000 clips (500 complete / 500 incomplete; 695 human / 305 TTS; 2.5 h), with median duration 9.1 s. For synthetic-to-human transfer, we additionally train on TTS speech only and evaluate on unseen human recordings. We evaluate conversational transfer zero-shot on the TurnBench dev set [20], which contains real two-channel human dialogue and uses event-anchored end-of-turn (EOT) evaluation. For this setting, Silero VAD [3] provides an acoustic gate and permits an EOT commit after at least 1 s of post-speech silence; FD-VAD receives no TurnBench training. We report evaluation results as mean of three random seed runs.

**Evaluation Metrics:** At the chunk level, we report precision, recall,  $F_1$ , and accuracy with STOP as the positive class. We separately report the *chunk false-stop rate*, defined as the fraction of chunks from

| Model                   | Chunk $F_1$  | Chunk Acc.   | False-stop $\downarrow$ |
|-------------------------|--------------|--------------|-------------------------|
| Always-CONTINUE         | 0.000        | 0.953        | 0.0%                    |
| WavLM-base causal [16]  | 0.183        | 0.721        | 29.1%                   |
| WavLM-large causal [16] | 0.191        | 0.724        | 29.0%                   |
| WavLM-large anchor [16] | 0.218        | 0.787        | 21.6%                   |
| Silero VAD [3]          | 0.251        | 0.790        | 22.5%                   |
| VAP [21]                | 0.279        | 0.803        | 20.1%                   |
| <b>FD-VAD</b>           | <b>0.832</b> | <b>0.984</b> | <b>0.6%</b>             |

**Table 1.** Chunk-level semantic endpoint detection on the held-out smart-turn-v3.1 [22] test set (1,000 utterances; 500 complete and 500 incomplete). STOP is the positive class; false-stop is the fraction of incomplete-speech chunks incorrectly predicted as STOP. FD-VAD outperforms streaming baselines with lowest false-stop rate.

|           | Model                   | Str. | Complete     | Incomplete   | Overall                |
|-----------|-------------------------|------|--------------|--------------|------------------------|
| ACOUSTIC  | Energy VAD              | ✓    | <b>0.998</b> | 0.680        | 0.839 [.82,.86]        |
|           | Silero VAD [3]          | ✓    | 0.996        | 0.196        | 0.596 [.56,.63]        |
| STREAMING | VAP [21]                | ✓    | 0.728        | 0.489        | 0.608 [.58,.64]        |
|           | WavLM-base causal [16]  | ✓    | 0.824        | 0.344        | 0.584 [.55,.62]        |
|           | WavLM-large causal [16] | ✓    | 0.904        | 0.264        | 0.584 [.55,.61]        |
|           | WavLM-large anchor [16] | ✓    | 0.908        | 0.268        | 0.588 [.56,.62]        |
|           | SoulX-Duplug [23]       | ✓    | 0.910        | 0.251        | 0.580 [.55,.61]        |
|           | X2-Turn [24]            | ✓    | 0.652        | 0.812        | 0.733 [.71,.76]        |
| OFFLINE   | Whisper+LLM             | ×    | 0.860        | 0.768        | 0.814 [.79,.84]        |
|           | Whisper+TEN [4]         | ×    | 0.940        | 0.660        | 0.800 [.77,.83]        |
|           | UltraVAD [25]           | ×    | 0.920        | 0.840        | 0.880 [.86,.90]        |
|           | Smart Turn [6]          | ×    | 0.966        | 0.966        | 0.965 [.96,.98]        |
|           | <b>FD-VAD</b>           | ✓    | 0.972        | <b>0.969</b> | <b>0.976</b> [.95,.98] |

**Table 2.** Utterance-level endpoint classification on the held-out smart-turn-v3.1 [22] test set with 95% bootstrap CI over 2K samples. FD-VAD shows the strongest balanced performance across complete and incomplete turns in streaming mode (Str), comparable to the strongest non-streaming semantic classifier.

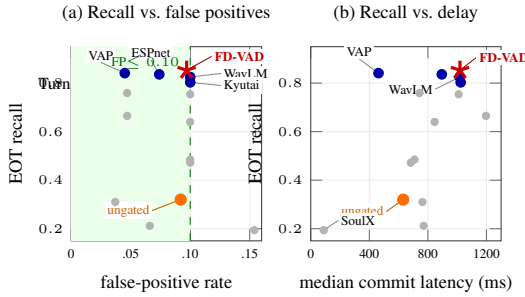
incomplete utterances incorrectly predicted as STOP. At the utterance level, we report endpoint accuracy separately for complete and incomplete queries, together with 95% bootstrap confidence intervals from 2000 resamples. We also report strict sentence accuracy (Sent-C/Sent-I), which requires all chunk decisions in an utterance to be correct, and near-boundary  $F_1$ , computed over predictions within  $\pm 2$  chunks of the true endpoint. For confidence-gated endpointing, the *false-interruption rate* is the fraction of incomplete utterances for which the system commits any premature STOP, and endpointing delay is  $L = t_{\text{commit}} - t_{\text{EOT}}$ , where  $L < 0$  denotes a commit before the annotated endpoint. We report this delay jointly with false-interruption rate. On TurnBench, we follow its EOT protocol and report recall at  $\text{FP} \leq 0.10$  together with median commit latency.

#### 5. RESULTS

**In-Domain Chunk and Utterance Performance:** Tables 1 and 2 show that FD-VAD reliably detects semantic endpoints while avoiding premature stops. At the chunk level, FD-VAD reaches  $F_1=0.832$  with only a 0.6% false-stop rate on incomplete utterances. Raw chunk accuracy is less informative because STOP accounts for only 4.8% of the labels; indeed, the always-CONTINUE baseline already achieves 0.953 accuracy. More importantly, the balanced utterance-level evaluation shows that FD-VAD performs consistently on both complete and incomplete turns (0.972/0.969), unlike acoustic VAD, whose performance drops sharply on incomplete speech (0.998  $\rightarrow$  0.680). This gap confirms that acoustic termination alone is insuffi-

| Category                  | System              | Recall       | FP                 | Lat. (ms) ↓ |
|---------------------------|---------------------|--------------|--------------------|-------------|
|                           | VAP [21]            | 0.841        | 0.045              | <b>463</b>  |
|                           | ESPnet turn-taking  | 0.836        | 0.074              | 895         |
|                           | ESPnet TT (per-ch.) | 0.640        | 0.100              | 846         |
|                           | Smart Turn [6]      | 0.754        | 0.100              | 1010        |
|                           | Moshi [1]           | 0.212        | 0.066              | 771         |
|                           | SoulX-Duplug [23]   | 0.194        | 0.154 <sup>†</sup> | 89          |
| Semantic                  | Kyutai semantic VAD | 0.803        | 0.100              | 1024        |
|                           | Gemini Live         | 0.665        | 0.047              | 1197        |
|                           | OpenAI semantic VAD | 0.310        | 0.037              | 763         |
|                           | OpenAI server VAD   | 0.933        | 0.563 <sup>†</sup> | 281         |
| Acoustic                  | WavLM-large anchor  | 0.825        | 0.100              | 1017        |
|                           | Mimi endpointer     | 0.759        | 0.047              | 742         |
|                           | WavLM-base causal   | 0.485        | 0.100              | 709         |
|                           | WavLM-large causal  | 0.472        | 0.100              | 683         |
|                           | RMS energy VAD      | 0.595        | 0.547 <sup>†</sup> | −98         |
| <b>Ours</b>               | <b>FD-VAD</b>       | <b>0.853</b> | <b>0.097</b>       | <b>1019</b> |
| <i>Oracle (annotator)</i> |                     | 1.000        | 0.000              | 0           |

**Table 3.** TurnBench [20] dev end-of-turn (EOT) performance under benchmark false-positive constraint ( $FP \leq 0.10$ ). FD-VAD is evaluated zero-shot and achieves the highest recall among qualifying systems. <sup>†</sup> denotes systems that exceed the false-positive budget.



**Fig. 3.** TurnBench [20] dev operating characteristics. Zero-shot gated FD-VAD achieves the highest EOT recall within the benchmark false-positive constraint ( $FP \leq 0.10$ ). The latency view highlights the complementary trade-off between endpoint reliability and commit speed; selected competitors are annotated and remaining baselines are shown in gray.

cient for semantic endpointing. FD-VAD also substantially outperforms the Whisper-based semantic cascades and performs relatively better compared to Smart Turn (0.965 vs. 0.976 overall), despite operating streaming rather than on a completed utterances.

**Zero-Shot Conversational EOT Transfer:** The in-domain evaluation uses isolated utterances, whereas practical endpointing operates within continuous dialogue. To apply FD-VAD to continuous audio, Silero VAD [3] provides an acoustic gate that determines *when* an endpoint is plausible, while FD-VAD determines *whether* the preceding speech is semantically complete. Table 3 shows that under the TurnBench false-positive constraint ( $FP \leq 0.10$ ), **FD-VAD achieves the highest EOT recall** among qualifying systems (0.853), despite being evaluated zero-shot. However, FD-VAD’s 1019 ms median commit latency is higher than VAP’s 463 ms. Fig. 3 further illustrates this trade-off: FD-VAD lies at the upper edge of the recall–false-positive operating region, but not on the best recall–latency frontier. The large separation between gated and ungated FD-VAD additionally shows that semantic endpoint prediction benefits substantially from acoustic gating in continuous dialogue.

**Streaming Latency Analysis:** A key advantage of FD-VAD over non-streaming ASR-based endpointing is that semantic inference

| Variant                 | Stride (ms) | Boundary sampling | Trainable adapter | Chunk Acc.   | Comp. $F_1$  | Near-b. $F_1$ | Sent-C Acc.  |
|-------------------------|-------------|-------------------|-------------------|--------------|--------------|---------------|--------------|
| FD-VAD (base)           | 320         | ×                 | ✓                 | 0.981        | 0.827        | 0.838         | 0.406        |
| 160-ms stride           | 160         | ×                 | ✓                 | 0.983        | 0.798        | 0.779         | 0.250        |
| Frozen adapter          | 320         | ×                 | ×                 | 0.978        | 0.803        | 0.816         | 0.378        |
| + Boundary sampling     | 320         | ✓                 | ✓                 | <b>0.987</b> | <b>0.838</b> | <b>0.880</b>  | <b>0.476</b> |
| Synthetic-only training | 320         | ✓                 | ✓                 | 0.972        | 0.785        | 0.786         | 0.341        |

**Table 4.** Ablation results on a matched 8k-clip subset. All variants use the same FD-VAD architecture and differ only in the listed settings. Near-b. is  $F_1$  within  $\pm 2$  chunks of the endpoint; Sent-C is strict all-chunk accuracy on complete utterances. We study robustness to training data by training FD-VAD on synthetic-only data that transfers well per chunk but degrades at utterance-level.

proceeds concurrently with the user’s speech rather than beginning only after the utterance completion. Let  $T_{\text{speech}}$  denote the utterance duration,  $T_{\text{stride}}$  is FD-VAD’s inference stride, and  $t_{\text{FD}}$  is its per-window processing time. An offline ASR-based cascade can produce its semantic endpoint decision only after the utterance has been observed, giving a endpoint decision time of  $T_{\text{speech}} + t_{\text{ASR}} + t_{\text{semantic}}$ , while FD-VAD continuously updates its endpoint state every  $T_{\text{stride}}$  ( $= 320$  ms) while speech is arriving. On a single NVIDIA A100, FD-VAD requires only 45.8 ms per 2.56 s causal window (46.8 ms p90), and this cost remains constant with utterance duration. At our operating point ( $\tau=0.9$ ,  $K=1$ ), the resulting median endpoint delay is 190 ms with a 2.0% false-interruption rate. Thus, longer utterances increase the latency of non-streaming semantic endpointing, whereas FD-VAD amortizes semantic reasoning through turns.

**Ablations and Stability:** Table 4 reveals two useful design trends. First, reducing the stride from 320 ms to 160 ms slightly raises aggregate chunk accuracy but substantially degrades complete-turn and near-boundary  $F_1$ . Second, jointly adapting the modality interface and explicitly emphasizing boundary examples are important: freezing the adapter degrades all endpoint-oriented metrics, whereas boundary-focused sampling yields the strongest complete-turn, near-boundary, and strict sentence performance. The three-seed variation is small ( $0.9804 \pm 0.0005$  chunk accuracy), indicating that these trends are not driven by a particular initialization.

**Robustness to Training Data Source:** FD-VAD model is trained on a mixture of human and synthetic speech. To assess its dependence on human training audio, we additionally train FD-VAD using only synthetic speech and evaluates it on held-out human recordings. The synthetic-only model retains similar per-chunk performance on human speech, with a 1.1-point reduction relative to mixed-data training, although the stricter all-chunk utterance metrics degrade more substantially, emphasizing the importance of human training data for consistent endpoint decisions across complete utterances.

## 6. CONCLUSION

We introduced FD-VAD, a streaming semantic endpoint detector that reformulates turn completion as causal audio-language reasoning over partial speech. By mapping bounded audio context directly to CONTINUE/STOP decisions, FD-VAD removes the need for intermediate transcription while remaining modular with respect to the downstream voice agent. Our evaluations show that FD-VAD transfers effectively to continuous real-time dialogue, where semantic reasoning complements conventional acoustic gating. We establish that direct audio-to-language reasoning provides a simple and portable foundation for turn-taking in full-duplex spoken agents.

## 7. REFERENCES

- [1] Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour, “Moshi: A speech-text foundation model for real-time dialogue,” Tech. Rep., Kyutai, 2024.
- [2] Xiong Wang, Yangze Li, Chaoyou Fu, Yike Zhang, Yunhang Shen, Lei Xie, Ke Li, Xing Sun, and Long Ma, “Freeze-omni: A smart and low latency speech-to-speech dialogue model with frozen LLM,” in *Proc. International Conference on Machine Learning (ICML)*, 2025, vol. 267 of *Proceedings of Machine Learning Research*, pp. 63345–63354.
- [3] Silero Team, “Silero VAD: Pre-trained enterprise-grade voice activity detector (vad), number detector and language classifier,” GitHub repository, 2024, Available: <https://github.com/snakers4/silero-vad>.
- [4] TEN Team, “TEN Turn Detection: Turn detection for full-duplex dialogue communication,” GitHub repository, 2025, Available: <https://github.com/TEN-framework/ten-turn-detection>.
- [5] Zhifu Gao, Shiliang Zhang, Ian McLoughlin, and Zhijie Yan, “Paraformer: Fast and accurate parallel transformer for non-autoregressive end-to-end speech recognition,” in *Proc. Interspeech*, 2022, pp. 2063–2067.
- [6] Pipecat AI, “Smart Turn v3: Native-audio semantic turn detection,” Open-source model and software, 2025, Available: <https://github.com/pipecat-ai/smart-turn>.
- [7] Guojian Li, Chengyou Wang, Hongfei Xue, Shuiyuan Wang, Dehui Gao, Zihan Zhang, Yuke Lin, Wenjie Li, Longshuai Xiao, Zhonghua Fu, and Lei Xie, “Easy turn: Integrating acoustic and linguistic modalities for robust turn-taking in full-duplex spoken dialogue systems,” *arXiv preprint arXiv:2509.23938*, 2025.
- [8] Chengyou Wang, Hongfei Xue, Chunjiang He, Jingbin Hu, Shuiyuan Wang, Bo Wu, Yuyu Ji, Jimeng Zheng, Ruofei Chen, Zhou Zhu, and Lei Xie, “Fastturn: Unifying acoustic and streaming semantic cues for low-latency and robust turn detection,” *arXiv preprint arXiv:2604.01897*, 2026.
- [9] Tristan Tsoi, Jiajun Deng, Yingke Zhu, Huu Quyen Dang, Tianxiang Cao, Nikita Kuzmin, Tao Zhong, and Simon Lui, “Next-turn: Duration-aware streaming endpoint detection via time-to-next-speech-onset prediction,” *arXiv preprint arXiv:2606.18094*, 2026.
- [10] Hyunjong Ok, Suho Yoo, and Jaeho Lee, “Speculative end-turn detector for efficient speech chatbot assistant,” in *Proc. 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, July 2026, pp. 45184–45197.
- [11] Yun-Sik Park and Sangmin Lee, “Voice activity detection using global speech absence probability based on teager energy for speech enhancement,” *IEICE Transactions on Information and Systems*, vol. E95-D, no. 10, pp. 2568–2571, Oct. 2012.
- [12] Zhiwei Lin, Tianjiao Du, Qiaochu Huang, Zihan Zhang, Naijun Zheng, Longshuai Xiao, Yunfei Lu, Jun Chen, and Zhiyong Wu, “TurnFSM for full-duplex dialogue system: Internalizing state-machine logic for streaming semantic voice activity detection and utterance-level rejection,” *arXiv preprint arXiv:2609.04240*, 2026.
- [13] Qingkai Fang, Shoutao Guo, Yan Zhou, Zhengrui Ma, Shaolei Zhang, and Yang Feng, “LLaMA-Omni: Seamless speech interaction with large language models,” in *Proc. International Conference on Learning Representations (ICLR)*, 2025.
- [14] Wenxi Chen, Ziyang Ma, Ruiqi Yan, Yuzhe Liang, Xiquan Li, Ruiyang Xu, Zhikang Niu, Yanqiao Zhu, Yifan Yang, Zhanxun Liu, Kai Yu, Yuxuan Hu, Jinyu Li, Yan Lu, Shujie Liu, and Xie Chen, “SLAM-Omni: Timbre-controllable voice interaction system with single-stage training,” in *Findings of the Association for Computational Linguistics: ACL 2025*, July 2025, pp. 2262–2282.
- [15] Yuhao Wang, Heyang Liu, Ziyang Cheng, Ronghua Wu, Qunshan Gu, Yanfeng Wang, and Yu Wang, “VocalNet: Speech LLMs with multi-token prediction for faster and high-quality generation,” in *Proc. Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Nov. 2025, pp. 19584–19601.
- [16] Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, Xiangzhan Yu, and Furu Wei, “WavLM: Large-scale self-supervised pre-training for full stack speech processing,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505–1518, Oct. 2022.
- [17] An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu, “Qwen2.5 technical report,” *arXiv preprint arXiv:2412.15115*, 2024.
- [18] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen, “LoRA: Low-rank adaptation of large language models,” in *Proc. International Conference on Learning Representations (ICLR)*, 2022.
- [19] Pipecat AI, “Smart Turn Data v3.1-train,” Hugging Face Dataset, 2025, Train split; accessed Sep. 1, 2026.
- [20] Freeman Jiang, Ramon Sanabria, Soham Deshmukh, Bandhav Veluri, Simon Michael Vuch Williams, Elliott K. Suen, Garreth Lee, Kevin Yoonho Choi, Takuya Umeki, Riku Kubo, Sathvik Udupa, Chien yu Huang, Shih-Yun Shan Kuan, Zhuoyan Tao, Satyapriya Krishna, Se-fik Emre Eskimez, Yu Tsao, Hung yi Lee, and Shinji Watanabe, “Turn-Bench: A multi-domain benchmark for turn-taking dynamics in spoken dialogue,” 2026.
- [21] Erik Ekstedt and Gabriel Skantze, “Voice activity projection: Self-supervised learning of turn-taking events,” in *Proc. Interspeech*, 2022, pp. 5190–5194.
- [22] Pipecat AI, “Smart Turn Data v3.1-test,” Hugging Face Dataset, 2025, Test split; accessed Sep. 1, 2026.
- [23] Ruiqi Yan, Wenxi Chen, Zhanxun Liu, Ziyang Ma, Haopeng Lin, Hanlin Wen, Hanke Xie, Jun Wu, Yuzhe Liang, Yuxiang Zhao, Pengchao Feng, Jiale Qian, Hao Meng, Yuhang Dai, Shunshun Yin, Ming Tao, Lei Xie, Kai Yu, Xinsheng Wang, and Xie Chen, “Soulx-duplug: Plug-and-play streaming state prediction module for realtime full-duplex speech conversation,” 2026.
- [24] Kaiqi Fu, Rime Wen, Altman Lin, Shawn Qin, Roy Gan, Hao Wang, and Qian Wang, “X2-turn: Frame-synchronous dual-head modeling for joint streaming asr and turn state prediction,” *arXiv preprint arXiv:2608.10878*, 2026.
- [25] Ultravox.ai Team, “UltraVAD: Context-Aware Audio-Native Neural Endpointing Model,” <https://huggingface.co/fixie-ai/ultraVAD>, 2025.